

Noesis Lab: 1–Page Report

1. Challenge Tackled

AI can quickly generate research ideas, but these ideas do not automatically result in faster discovery. AI research systems often waste time by testing topics that are already known, following noisy results, or repeating experiments without iteratively learning what to attempt next. *Noesis Lab* helps ML researchers and AI teams decide what's actually worth researching. It connects prior research, hypothesis evaluation, real experiments, and the next research decision in one automated loop.

2. Tools / ML Models Used

- **Claude Opus 5.5 / Anthropic API** — Literature, Scientist, and Critic agents.
- **Python, PyTorch + Apple MPS, TinyShakespeare** — local small-language-model experiment testbed.
- **arXiv** — curated literature snapshot of 10 papers and 13 extracted claims.
- **Pydantic + SQLite** — structured agent outputs and experiment/evidence storage.
- **Streamlit, NumPy, pandas** — dashboard, analysis, and result processing.

3. What Worked Well

- Built a complete loop from **literature** → **hypothesis** → **experiment** → **measured result** → **next decision**, rather than only generating research ideas.
- The Literature Agent can identify known ideas before compute is spent and link its decision to an exact passage from a real paper.
- AI proposes and interprets, but **deterministic code controls measurements and experimental decisions**, preventing the model from inventing/hallucinating results.
- **Final measured result (explore3 notebook)**: 20 runs across 9 hypotheses, 3 avoided by prior-art detection. Pre-LN rejected as known, RMSNorm soft-rejected as covered. ReLU²+RoPE held on every paired seed at 2.35× noise SD and was promoted to the new baseline. QK-norm + no warmup flagged harmful at 2.5× noise SD.

4. What Was Challenging

- **Separating reasoning from evidence**: LLMs cannot create measurements or determine experimental success; only executed runs and fixed rules can.
- **Handling experimental noise**: we measured a five-seed baseline and compared candidates on paired seeds, clearly labeling results as screening rather than confirmed findings.
- **Grounding citations**: literature claims use stored passages from a curated arXiv snapshot rather than model-generated quotations.
- **Reproducibility**: agent outputs and experiment results can be recorded/replayed, and statistics can be recomputed from stored runs.

5. How We Spent Our Time

- **10:00 AM–12:30 PM**: Kickoff, research, repeated specification refinement, and scope decisions.
- **12:30–5:00 PM**: Built agents, literature system, experiment harness, decision logic, storage, dashboard.
- **7:00–10:00 PM**: Testing, integration, and debugging.
- **11:30 PM–3:30 AM**: Final validation, experiment runs, documentation, videos, and submission.

If we had 24 more hours, we would expand the literature corpus, utilize more agents, run more confirmation experiments, and build a more polished researcher-facing interface.